ÁRKTALAŠ HOLVVAS

# ARCTIC SCREAM




## VISUAL MANUAL · 1.5.2
## FOR THE M-VAVE FM-1

Dark ambient drones and feedback screams from the arctic tundra.
Unofficial firmware by PolarPatch, free and open source (GPL-3.0); not made or endorsed by M-VAVE. Install at your own risk.

00

# How to read this manual

Every page of Arctic Scream is shown as the FM-1's screen shows it, taken from the firmware itself, with a table of what its knobs and buttons do. The four tiles along the bottom of the screen always belong to KNOB 1–4, left to right, each in its colour.

## Words used here

- **Tap** a button: open its page. Tap again: its next page.
- **Hold** a button: its second job (written as *hold*).
- **A + B**: hold A, then press B.
- **Keys from F2:** everything sits an octave down, for drones. **White keys 1–8** are counted from the left (F2, G2, A2, B2, C3, D3, E3, F3); the black keys from **F#2**.
- **T1–T4:** the four tracks. Turn ALGORITHM to pick one; most pages show the selected track.
- NEW marks what is new in 1.5, 1.5.1 and 1.5.2.

## Arctic Scream p0larpatch

by PolarPatch




A drone firmware for the M-VAVE FM-1: four tracks, eight sources, long dark effects, a 64-step sequencer, chords, an arpeggiator and song mode, with its own feedback voice, Holvut, that wakes on its own and screams.

Free and open source (GPL-3.0). Try it in your browser first: the demo, the editor and the cheat sheet are at polarpatch.github.io/Arctic-scream-FM-1.

01

# Getting around

Four tracks play together and share six voices. The buttons open the pages, the four knobs set the four tiles at the bottom of the screen, and these controls work everywhere. Every track starts with ICE WRAITH: press a low key and let it howl.

## Every page

No button needed

| Control | What it does | Range |
|---|---|---|
| **ALGORITHM** | Picks the track. Every page shows that track. | T1–T4 |
| **PRESETS** | Loads the next or previous sound into the track (factory first, then yours). On the sequencer pages: the tempo. | 47 factory + 64 yours |
| **SELECT** | On HOME: the tempo. On the sequencer: the cursor. | 20–300 BPM |
| **MASTER** | Volume. | |
| **OCT– / OCT+** | Octave down and up. In the save menus: back and confirm. | |
| **Keys** | Play the selected track, snapped to the SCALE (chord mode: its chord; ARP on: arpeggiated). | F2 and up |
| **LATCH ON: keys** | A key starts its note, the same key again stops it. | |
| **PLAY** | Starts and stops all four sequences (following a MIDI clock: at its next pulse). | |
| **REC** | Record on or off. On SONG: arms song recording. | |
| **REC: hold ~1 s** | Clears the selected track's pattern (undoable). | |
| **EDIT + OCT– / OCT+** | Undo / redo the last edit. | |
| **SEL + OCT+** | The selected track's LATCH on or off, from any page (off lets go of its latched notes). | |
| **Key lights** | Latched and arpeggiated notes stay lit, and their lights follow OCT– and OCT+. | |
| **OCT– + OCT+** | Sets SYSTEM 2's CPU PEAK and USB REC DROPS to 0. | |

> Saving is automatic: stopped and silent, your work saves itself and comes back at power-on. Under a latched drone it saves after 30 s left alone; a slot or a sound you save is written at once (a short gap in the sound). Knobs speed up: slowly for fine steps, fast to sweep.

## Buttons: tap and hold

What each button opens

**HOME**
HOME → MIXER 1 → MIXER 2. Hold: SYSTEM (then HOME: SYSTEM 2, SYSTEM 3).

**EDIT**
SOUND → SOURCE (FM: SOUND → FM ALGO → FM OP → SOURCE; Holvut: SOUND → LOOP → SOURCE).

**ENV**
ENV.

**LFO**
LFO → DRONE.

**SEL**
SCALE → CHORD. Hold and let go: chord mode on or off.

**ARP**
ARP. Hold: the arpeggiator on or off for the track.

**SEQ**
GRID → PATTERN SETTINGS. Hold + white keys: sequencer tools; hold + SELECT: pattern length.

**SAVE**
SOUNDS → PROJECTS. Hold + white keys 1–4: recall slots A–D, 5–8: save into them.

**GLO**
SONG. Hold + white keys 1–4: mute T1–T4, 5–8: solo them.

**FX**
TAPE → GRAIN → DELAY → SPACE → MASTER → DRIVE → AIR. Hold: PERFORM (+ black keys: GENERATE).

02

# HOME and the mixers

Each track's wave glows on HOME as it plays, under a starry sky. Tap HOME again for the mixers: MIXER 1 draws each track as a curtain of standing ribbons, MIXER 2 as a ribbon across the stereo field.

## HOME

Tap HOME




| Control | What it does | Range |
|---|---|---|
| KNOB 1 VOLUME | The selected track's level (MUTE when muted). | 0–100 % |
| KNOB 2 CUTOFF | The track filter's cutoff (MIXER 2's TYPE), for every source. | ~30 Hz – OPEN |
| KNOB 3 WARP | Eight layers of motion that come in one after another: the voices drift apart, the filter breathes, the knobs wander, the storm pushes, vibrato, resonance, wide drift, at the top flutter and a sagging pitch. Smooth from still to wild; the level held. | 0–100 % |
| KNOB 4 MORPH | One knob through all four of the source's knobs at once (the stepped ones, such as MODEL, stay), on a path of its own, wandering a little around it. At 0 the sound is as set. | 0–100 % |
| SELECT | The tempo (following a MIDI clock: EXT and its tempo). | 20–300 BPM |

## MIXER 1

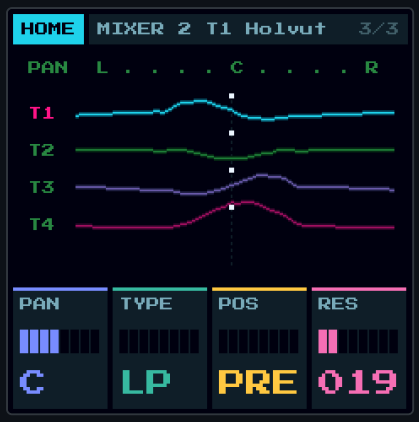
Tap HOME twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1 LEVEL | The track's level. | 0–100 % |
| KNOB 2 DLY SEND | Send to DELAY. | 0–100 % |
| KNOB 3 REV SEND | Send to SPACE. | 0–100 % |
| KNOB 4 GRAIN SEND | Send to GRAIN. | 0–100 % |

# MIXER 2

Tap HOME three times




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>PAN | Where the track sits. | L64 – C – R63 |
| KNOB 2<br>TYPE | The track filter: low-pass, high-pass or band-pass. LP open (CUTOFF at the top) or HP open (at the bottom): no filtering. BP is gentle at RESONANCE 0 and narrows as you turn it up. | LP / HP / BP |
| KNOB 3<br>POS | PRE: the effect sends hear the filtered sound. POST: only the dry sound is filtered; GRAIN, DELAY and SPACE hear it unfiltered (a dark track with a bright reverb). | PRE / POST |
| KNOB 4<br>RESONANCE | The track filter's resonance (on HOME before 1.3). | 0–100 % |

03

# Sources: EDIT

EDIT opens the track's SOUND page, the source's four knobs, then SOURCE (a Holvut track has LOOP between them). Eight sources, each with its own picture.

## SOURCE

Tap EDIT twice (Holvut: three times)




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>SOURCE | Picks the track's source. | VA, FM, SAMPLE, VOICE, WHEEL, PHYS, PHASE, Holvut |
| KNOB 2<br>LATCH | Holds the notes: a key starts its note, the same key again stops it. | OFF / ON |

## Holvut, the feedback voice

EDIT, source Holvut



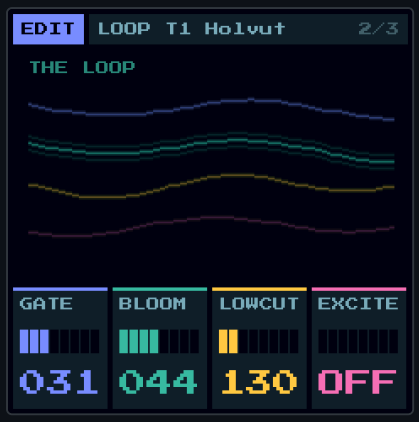

| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>FEEDBACK | Wakes the loop. Below the threshold it sleeps (THE LOOP: ASLEEP); past ~2/3 it grows out of silence and howls by itself (HOWLING). Its last steps (120–127) overblow: the howl leaps up the overtones. | 0–100 % |
| KNOB 2<br>BODY | Bends it: the resonant body, ~1 to ~92 ms. Past 100 slow beating howls, past 115 slow pulses. | 0–100 % |
| KNOB 3<br>DAMP | Turns the scream into a growl. | 0–100 % |
| KNOB 4<br>DRIVE | Tears it. | 0–100 % |

Holvut is North Sami for to howl. A string and a resonant body sit inside a loop with saturation, filters and diffusion; nothing is struck. Each key sends a burst into the loop, so a note near the threshold sets it off.

Pressure: from a MIDI keyboard with aftertouch, pressing harder feeds the note back more, then overblows.

## LOOP

EDIT twice on a Holvut track




| Control | What it does | Range |
|---|---|---|
| KNOB 1 GATE | When the key is let go, the loop stops feeding itself: all the way up it dies into its own tail, in between it fades but lingers. | OFF, 1–100 % |
| KNOB 2 BLOOM | What GRAIN, DELAY and SPACE add to the mix goes back into the loop: an FX loop inside the feedback. Turn up the track's sends. | OFF, 1–100 % |
| KNOB 3 LOWCUT | A high-pass inside the loop, two poles: the howl thins and climbs; with DAMP, a band that screams. | ~70 Hz – ~1.9 kHz |
| KNOB 4 EXCITE | The track's sample (SAMPLE's SLOT, START, LENGTH, looped) fed into the string: your own sound wakes the howl. The tile shows its slot (S2 45%) or EMPTY. | OFF, 1–100 % |

> All four at OFF sound as before 1.4. Saved with the sound and the project.

> Each knob is a ribbon: GATE's tail fades, BLOOM's echoes appear around it, LOWCUT takes the swell out of it, EXCITE throws sparks along it.

## VA

EDIT, source VA




| Control | What it does | Range |
|---|---|---|
| KNOB 1 WAVE | Virtual analog: the waveform. | SAW / SQR / SUPER |
| KNOB 2 DETUNE | Detunes the voices against each other. | 0–100 % |
| KNOB 3 CUTOFF | The source's own filter. | 0–100 % |
| KNOB 4 RESONANCE | Its resonance. | 0–100 % |

## FM: SOUND, FM ALGO, FM OP

EDIT on an FM track: SOUND → FM ALGO → FM OP




| Control | What it does | Range |
|---|---|---|
| SOUND · KNOB 1 | PATCH: the six-operator patch. | |
| SOUND · KNOB 2–4 | BRIGHT, DECAY, DETUNE. | 0–100 % |
| ALGO · KNOB 1 | ALGO: how the six operators are wired. | 01–32 |
| ALGO · KNOB 2 | FEEDBACK. | |
| ALGO · KNOB 3 | OP: the operator the next page edits. | 1–6 |
| ALGO · KNOB 4 | FINE: the operator's ratio, fine. | |
| OP · KNOB 1–4 | RATIO, LEVEL, ATTACK, DECAY of the operator. | LEVEL etc. 0–99 |

> The screen draws the wiring: carriers (the ones you hear) outlined, modulators filled, the selected one lit.

## SAMPLE

EDIT, source SAMPLE




| Control | What it does | Range |
|---|---|---|
| KNOB 1 SLOT | One of the four sample slots, played as a loop. | 1–4 |
| KNOB 2 START | Where the loop starts. | 0–100 % |
| KNOB 3 LENGTH | How long it is. | |
| KNOB 4 REVERSE | Backwards. | OFF / ON |

> No samples are built in: the four slots are yours, filled with WAVs from the editor. An empty slot is silent.

## VOICE

EDIT, source VOICE




| Control | What it does | Range |
|---|---|---|
| KNOB 1 VOWEL | A singing formant voice: the vowel. | A E I O U |
| KNOB 2 SHIFT | Shifts the formants. | |
| KNOB 3 BUZZ | The buzz of the throat. | 0–100 % |
| KNOB 4 BREATH | Air in it. | 0–100 % |

## WHEEL

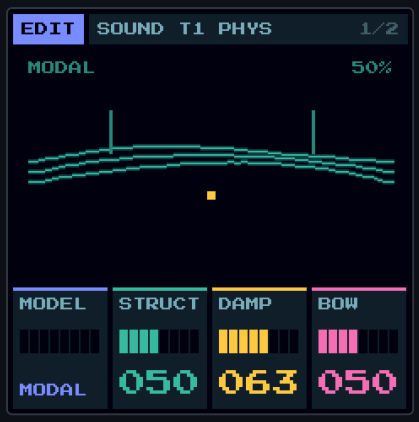
EDIT, source WHEEL




| Control | What it does | Range |
|---|---|---|
| KNOB 1 DRAWBARS | A tonewheel organ: the registration. | 16 settings |
| KNOB 2 PERC | Percussion. | OFF and 6 kinds |
| KNOB 3 CLICK | The key click. | 0–100 % |
| KNOB 4 DRIVE | Overdrive. | 0–100 % |

## PHYS

EDIT, source PHYS




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>MODEL | Bowed and struck strings, bars, plates, membranes (MEMB), sympathetic strings (SYMP). | |
| KNOB 2<br>STRUCTURE | The body's structure (MEMB: HARM, SYMP: CHORD). | 0–100 % |
| KNOB 3<br>DAMP | How fast it dies away. | 0–100 % |
| KNOB 4<br>BOW | Bowing (MEMB: BEND, SYMP: BUZZ). | 0–100 % |

> The names of KNOB 2 and 4 change with the model; the screen shows the current one.

## PHASE

EDIT, source PHASE




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>WAVE | Phase distortion: the wave. | |
| KNOB 2<br>WAVE2 | The second cycle's wave. | |
| KNOB 3<br>DCW | How far it is bent. | 0–100 % |
| KNOB 4<br>DETUNE | Detune. | 0–100 % |

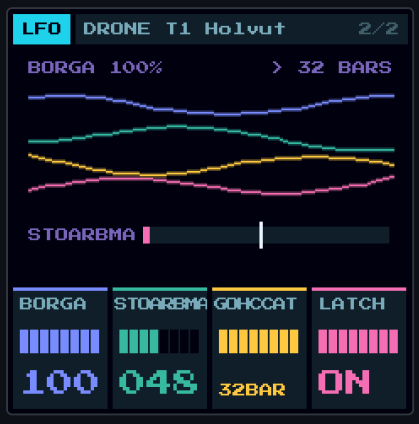
04

# DRONE: a sound that lives on its own

Press LFO twice. DRONE makes a held sound move by itself, slowly, and never the same way twice. The knobs have North Sami names.

## DRONE

Tap LFO twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1 **BORGA (snowdrift)** | Four slow random walks move the filter, the source's shape, its most natural knob (on Holvut: FEEDBACK, around the threshold) and the reverb's distance. Each voice drifts a few cents and breathes: a held chord sounds like a choir. | 0–100 % |
| KNOB 2 **STOARBMA (storm)** | The tension: opens the filter, raises resonance and drive, spreads the voices, makes the walks deeper and faster. | 0–100 % |
| KNOB 3 **GOHCCÁT (to wake)** | How long the storm takes to rise, in tempo. A drone woken from silence starts calm and builds. | OFF, 1–32 bars |
| KNOB 4 **LATCH** | Holds the notes. | OFF / ON |

> At 0 % nothing is added. The settings are saved with the sound; the seven Holvut presets start with BORGA on.

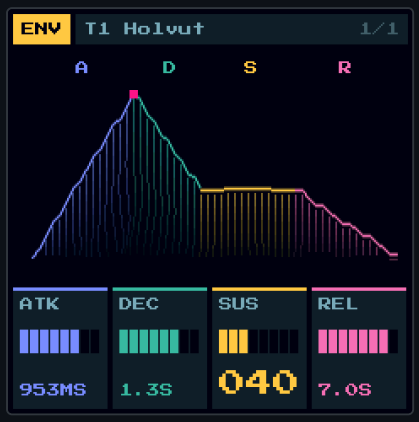
05

# ENV and LFO

How each note starts and ends, and slow movement. Both belong to the track.

## ENV

Tap ENV




| Control | What it does | Range |
|---|---|---|
| KNOB 1 ATTACK | How each note starts. | ~1 ms – 10 s |
| KNOB 2 DECAY | How fast it falls to SUSTAIN. | ~10 ms – 10 s |
| KNOB 3 SUSTAIN | The level it holds. | 0–100 % |
| KNOB 4 RELEASE | How it dies away after the key. | ~10 ms – 20 s |

## LFO

Tap LFO




| Control | What it does | Range |
|---|---|---|
| KNOB 1 RATE | How fast. It never restarts with a note, so drones keep breathing. | 60 s a cycle – 20 Hz |
| KNOB 2 DEPTH | How much (0 = off). | 0–100 |
| KNOB 3 DESTINATION | What it moves. VOLUME swings around the track's VOLUME, so a track at 0 still sounds on its peaks: mute it with GLO + keys 1–4. | PITCH, VOLUME, PAN, the source's four knobs, REV SEND, GRAIN SEND, CUTOFF |
| KNOB 4 SHAPE | Its shape. | SINE, TRIANGLE, SAW UP, SAW DOWN, SQUARE, RANDOM, DRIFT |

06

# Scales, chords and the arpeggiator

SEL opens SCALE and CHORD; hold SEL and let go for chord mode: one key, a whole chord. ARP opens the arpeggiator.

## SCALE

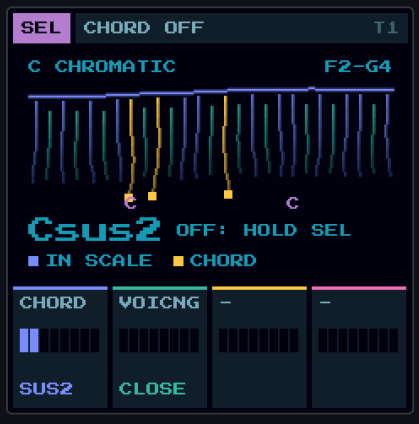
Tap SEL




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>ROOT | The scale's root, shared by all tracks. | C–B |
| KNOB 2<br>SCALE | Keys, MIDI and recordings snap to it. The last is HORROR: close semitones, every note wrong in the right way. | OFF + 14 scales |

## CHORD

Tap SEL twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>CHORD TYPE | The chord a key plays in chord mode. | SCALE, MIN, MAJ, SUS2, SUS4, FIFTH, MIN7, MAJ7, DIM, ADD9, CLUSTER |
| KNOB 2<br>VOICING | How it is spread. Big chords use more of the six voices. | CLOSE, OPEN, SPREAD, +OCT, DROP |

## ARP

Tap ARP · hold ARP: on / off




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>MODE | The order it plays the held keys in. | UP, DOWN, UP-DOWN, RANDOM, AS PLAYED, CHORD |
| KNOB 2<br>RATE | How fast, on the sequencer's clock. | 1/1 – 1/32, triplets |
| KNOB 3<br>OCTAVES | How many octaves it climbs. | 1–4 |
| KNOB 4<br>GATE | How long each note sounds. | 10–100 % |

> It plays the keys you hold; with LATCH ON it keeps going after you let go.

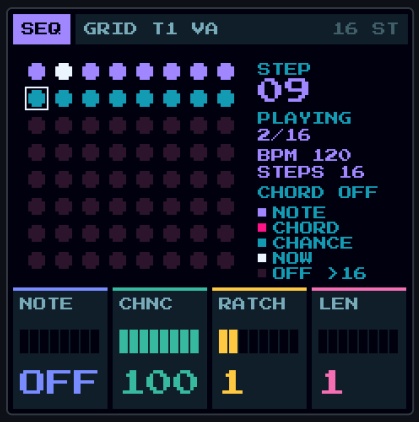
07

# Sequencer: SEQ, PLAY, REC

Each track has its own 64-step pattern; all four play together on one tempo. Stopped + REC: each key fills a step and moves on. Playing + REC: notes land on the nearest step.

## GRID

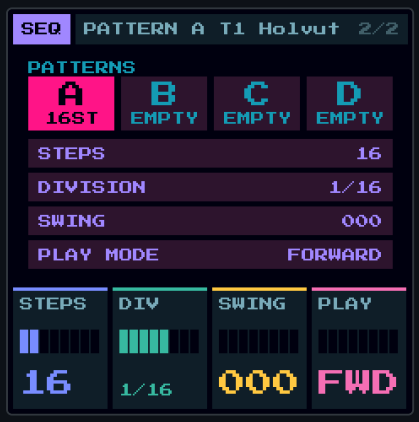
Tap SEQ




| Control | What it does | Range |
|---|---|---|
| KNOB 1 NOTE | The selected step's note (all the way down: OFF). | OFF, notes |
| KNOB 2 CHANCE | How likely it plays. | 0–100 % |
| KNOB 3 RATCHET | Repeats within the step. | 1–4 |
| KNOB 4 LENGTH | How long it sounds. | |
| SELECT | Moves the cursor. PRESETS: the tempo. | |
| SEL + a knob | Locks that value to the step. SEL + OCT–: clears the step's locks. | |

## PATTERN SETTINGS

Tap SEQ twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1 STEPS | The pattern's length (also: hold SEQ + turn SELECT). | 1–64 |
| KNOB 2 DIVISION | How long a step is. | 1/16 etc. |
| KNOB 3 SWING | Swing. | 0–100 % |
| KNOB 4 PLAY MODE | Which way it runs. | FORWARD and more |

**Hold SEQ + a white key** (all undoable):

| F2 | G2 | A2 | B2 | C3 | D3 | E3 | F3 |
|---|---|---|---|---|---|---|---|
| CLEAR | RANDOM | SHIFT ← | SHIFT → | COPY | PASTE | DOUBLE | MUTATE |

08

# Sounds, slots and songs: SAVE, GLO

A slot (A–D) holds the four tracks' sounds, sequences, locks and mix. Hold SAVE + white keys 1–4 to recall them (at the next bar while playing), 5–8 to save into them (press again within 3 s to overwrite).

## SOUNDS

Tap SAVE




| Control | What it does | Range |
|---|---|---|
| KNOB 1 SLOT | One of 64 slots for your own sounds. | 001–064 |
| KNOB 2 CATEGORY | Its category: it comes in PRESETS after that category's factory sounds. | |
| KNOB 3 TYPE | Shows only one source's sounds. | ALL, the sources |
| KNOB 4 ACTION | Confirm with OCT+, back out with OCT–. | LOAD / SAVE / CLEAR |

A sound keeps its source and knobs, ENV, LFO, DRONE, LATCH, chord and (FM) its patch. Name it in the editor (up to 10 characters).

## PROJECTS

Tap SAVE twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1 SLOT | One of the four slots. | A–D |
| KNOB 2 ACTION | Confirm with OCT+. | LOAD, SAVE, ERASE, NEW |

# SONG

Tap GLO




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>SECTION | A step of the chain. | 01–16 |
| KNOB 2<br>PATTERN | The slot it plays (END: the song ends there). | A–D / END |
| KNOB 3<br>REPEATS | How many times. | 01–16 |
| KNOB 4<br>MODE | OFF: PLAY loops the slot. SONG: once. SONG LOOP: over and over. | OFF / SONG / LOOP |

> Record a song: on SONG press REC, then PLAY, and switch slots with SAVE + keys 1–4.

> Hold GLO + white keys 1–4 mutes T1–T4, keys 5–8 solos them.

09

# Destroy: FX

Tracks feed GRAIN, DELAY and SPACE through their sends; MASTER, DRIVE and TAPE work on the whole mix. Each page has its own northern lights.

## TAPE

Tap FX

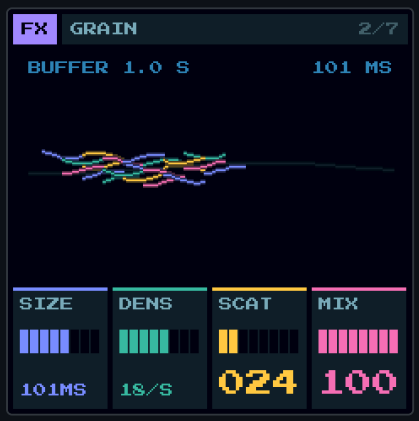



| Control | What it does | Range |
|---|---|---|
| KNOB 1 WOW | Slow wobble and flutter, up to a seasick ±2 % of pitch. | 0–100 % |
| KNOB 2 SATURATION | Tape saturation, up to 12x drive. | 0–100 % |
| KNOB 3 TONE NEW | Darker as it turns up, down to ~400 Hz (DRIVE's TONE goes the other way). | OPEN, 10K … 400 |
| KNOB 4 HISS | Tape hiss, up to loud. | 0–100 % |

All four at 0 switch TAPE off (BYPASS). It colours the whole mix at the very end.

## GRAIN

Tap FX twice



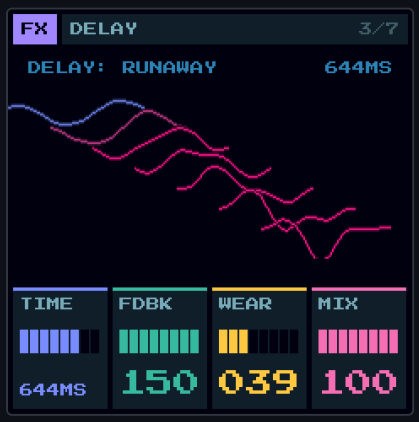

| Control | What it does | Range |
|---|---|---|
| KNOB 1 SIZE | How long each grain is. | 10–390 ms |
| KNOB 2 DENSITY | How many grains a second. | |
| KNOB 3 SCATTER | How far they scatter. | 0–100 % |
| KNOB 4 MIX | How loud the cloud is. | 0–100 % |

FREEZE (PERFORM, F2) holds the grain cloud. GRAIN pours into SPACE.

GRAIN hears what the tracks send it (MIXER 1). Turn a knob up while the selected track sends nothing and its send goes to 50 %.

## DELAY

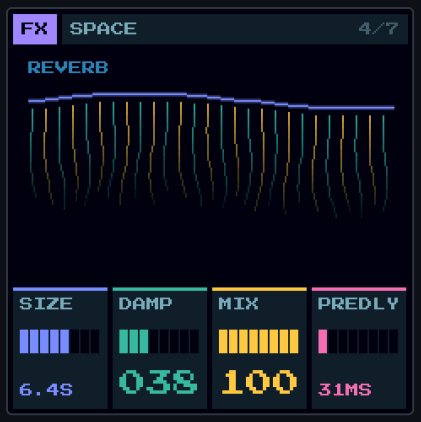
Tap FX three times




| Control | What it does | Range |
|---|---|---|
| KNOB 1 TIME | The echo's time. | 20 ms – 2.6 s |
| KNOB 2 FEEDBACK | Past 100 % the echo grows and saturates, and burns magenta on the screen: runaway. | 0–150 % |
| KNOB 3 WEAR | Worn, wobbly echoes: darker, driven to 4x, wobbling up to ~11 ms. | 0–100 % |
| KNOB 4 MIX | How loud they are. | 0–100 % |

## SPACE

Tap FX four times




| Control | What it does | Range |
|---|---|---|
| KNOB 1 SIZE | The reverb's length; at the top it holds for ever (INF, FREEZE). | to ~32 s, INF |
| KNOB 2 DAMP | Darkens the tail. | 0–100 % |
| KNOB 3 MIX | How loud it is. | 0–100 % |
| KNOB 4 PRE-DELAY | The gap before it. | 0–250 ms |

## MASTER

Tap FX five times




| Control | What it does | Range |
|---|---|---|
| KNOB 1 CUTOFF | A filter on everything, in stereo. LP at the top or HP at the bottom: off completely (OPEN). | ~30 Hz – OPEN |
| KNOB 2 RESONANCE | Its peak at the cutoff: the rays there flare. | 0–100 % |
| KNOB 3 TYPE | Low-pass, high-pass, band-pass. | LP / HP / BP |
| KNOB 4 POSITION | PRE FX: the tracks' dry mix before the effects return (reverb and echoes stay bright). POST REVERB: tracks and effects together, before DRIVE and TAPE. END: the very end, after TAPE and PERFORM. | PRE FX / POST REVERB / END |

The picture: an aurora curtain cut by the filter, low on the left, high on the right.

## DRIVE

Tap FX six times



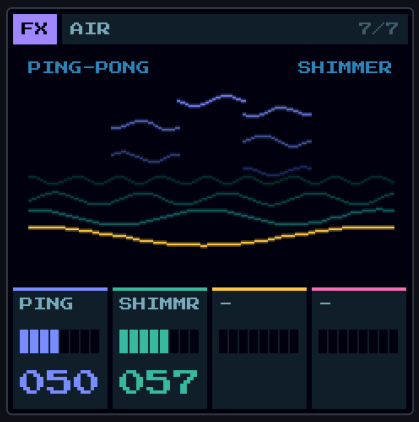

| Control | What it does | Range |
|---|---|---|
| KNOB 1 AMOUNT | How hard it tears the whole mix, after SPACE, before TAPE. 0: off completely. | OFF, 1–100 % |
| KNOB 2 TYPE | SOFT: warm saturation. FUZZ: lopsided, buzzy. CLIP: hard. FOLD: a wavefolder, folding back into new overtones. | SOFT / FUZZ / CLIP / FOLD |
| KNOB 3 TONE | Darkens it. | 0–100 %, OPEN |
| KNOB 4 MIX | Blends it with the clean sound. | 0–100 % |

All four types are about equally loud at the same AMOUNT; all the way up is at most ~6 dB louder, and a limiter after it keeps the output safe. Turn AMOUNT on KNOB 1 for fast changes.

MASTER and DRIVE are saved with the project, like the other effects.

# AIR

Tap FX seven times




| Control | What it does | Range |
|---|---|---|
| KNOB 1<br>PING | PING-PONG: DELAY's echoes thrown left, right, left, wider as you turn it. While on, TIME goes up to 1.3 s. | OFF, 1–100 % |
| KNOB 2<br>SHIMMR | SHIMMER: SPACE's tail taken an octave up and back into the reverb, so it rises as it fades. | OFF, 1–100 % |

The picture: DELAY's echoes flying apart above, SHIMMER's octaves rising over SPACE's tail below.

Saved with the project; older projects load with both off.

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# Perform and GENERATE: hold FX

Hold FX for a moment and PERFORM appears. Each trick lasts while you hold its white key; the sequencer keeps running, and you land back in time when you let go. The black keys start GENERATE.

## PERFORM

Hold FX + white keys




| F2 | G2 | A2 | B2 | C3 | D3 | E3 | F3 |
|---|---|---|---|---|---|---|---|
| FREEZE | REV THROW | ECHO THROW | TAPE STOP | REVERSE | FILTER DOWN | FILTER UP | STUTTER |

FREEZE is a toggle: tap it to hold the grain cloud, tap again to let go.

## GENERATE

Hold FX + black keys




| Control | What it does | Range |
|---|---|---|
| F#2 G#2 A#2 C#3 | Tapped: GENERATE on or off for T1–T4. The track plays by itself: random notes of the SCALE (chord mode: chords), on the tempo, resting now and then. | |
| D#3 | MODE. DRIFT: few long, soft notes. PULSE: a melody on eighths. STORM: dense sixteenths, wild volume. | DRIFT / PULSE / STORM |
| F#3 / G#3 | OCTAVE − / +: the generator's own (0: from C3). | −3 … +3 |
| A#3 | RANGE: how far up it may play. | 1, 2, 3 OCT |

| F#2 | G#2 | A#2 | C#3 | D#3 | F#3 | G#3 | A#3 |
|---|---|---|---|---|---|---|---|
| T1 | T2 | T3 | T4 | MODE | OCT − | OCT + | RANGE |

With a track's key held, D#3–A#3 set only that track; alone, all four.

The PERFORM screen shows a tile a track: lit while it generates, its MODE, its range, and a square that flashes with each note. Its own pattern runs on silently underneath and comes back when you switch GENERATE off. Nothing is saved: after power-on it is off.

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# System: hold HOME

Hold HOME for SYSTEM; press HOME again for SYSTEM 2, again for SYSTEM 3. These settings belong to the FM-1, not the project.

## SYSTEM

Hold HOME




| Control | What it does | Range |
|---|---|---|
| KNOB 1 COLOUR | Pick the one that matches your FM-1. | ICE, AURORA, VIOLET, PLASMA |
| KNOB 2 MIDI CHANNEL | 1-4: T1–T4 on channels 1–4 (or higher, to 13-16). ALL > SEL: everything to the selected track. CH n > SEL: only that channel, to the selected track. | 1-4 … 13-16, ALL, CH 1–16 |
| KNOB 3 TUNE | Fine tuning. | ±50 cents |
| KNOB 4 SCREEN OFF | The screen goes dark after this; the music goes on. The first touch only wakes it. | 1–… min, NEVER |

## SYSTEM 2

Hold HOME, then HOME




| Control | What it does | Range |
|---|---|---|
| KNOB 1 MIDI LEARN | Set LEARN, turn a knob on the FM-1, then move one on your controller. Up to 32 links. | OFF, LEARN, CLEAR |
| KNOB 2 SEQ MIDI OUT | Sends the sequencer's and the arpeggiator's notes out over USB. | OFF / ON |
| KNOB 3 SAMPLES | CLEAR and OCT+ empties all four sample slots. | KEEP / CLEAR |
| KNOB 4 CLOCK | INT: its own tempo. I+OUT: the same, and MIDI clock out over USB. USB or TRS: follows a MIDI clock from there (START, CONTINUE, STOP). | INT, I+OUT, USB, TRS |

> The processor load, NOW and PEAK, and USB REC DROPS: gaps in the sound sent to a computer. OCT– + OCT+ sets them to 0.

## SYSTEM 3

Hold HOME, then HOME twice




| Control | What it does | Range |
|---|---|---|
| KNOB 1 LATCH | FROM SOUND: a sound you load brings its own LATCH (the seven Holvut sounds latch). ALWAYS OFF: loading sets it off; EDIT's LATCH and SEL + OCT+ still switch it. | FROM SOUND / ALWAYS OFF |
| KNOB 2 USB LEVEL | How loud a computer records the FM-1. 0 DB: the volume knob all the way up; higher makes quiet drones louder in the recording, a limiter keeps them from clipping. | –12 … +24 DB |
| KNOB 3 MOD NEW | MOD OUT: each track's level (CC 102) and LFO (CC 103) out over USB 50 times a second, on the track's channel. For lights, motors, Max or another synth. | OFF / ON |

# MIDI, USB audio and good to know

| Control | What it does |
|---|---|
| MIDI in | USB (the MIDI port is called **Arctic Scream** too), or the TRS jack with a TRS-to-DIN adapter (type A). Notes play the tracks, a CC its MIDI LEARN link; ALL NOTES OFF lets go of MIDI's notes; aftertouch pushes Holvut's FEEDBACK. |
| USB audio | Class compliant, no driver: the FM-1 shows up on the computer, phone or tablet as **Arctic Scream**, an audio input, 16-bit stereo at 44.1 or 48 kHz, the whole mix at SYSTEM 3's USB LEVEL (the volume knob sets the speaker, not the recording). Solo each track (GLO + keys 5–8) to record them one by one. |
| Headphones | On wide, loud sound the FM-1's headphone amplifier used to crackle; a limiter now holds the difference between left and right in check near MASTER's top. Mono sound and the USB recording are untouched. |
| CPU FULL: NOTE DROPPED | It had more to play than it can: a fading or generated note goes first, your held notes last. |
| WATCHDOG / LOW VOLTAGE RESET | Why it restarted: it stood still, or the power dipped. |
| Screen reader NEW | The FM-1 says its screen in words: open **charlesvestal.github.io/fm1-reader** in Chrome or Edge and connect. Pages, knobs as they turn, lists, steps and messages, in its own voice or through your screen reader. |
| Saves NEW | A slot or a sound you save is written at once, a drone sounding or not; the automatic save goes through after 30 s left alone under a latched drone. The editor keeps a project's effects and tempo in its file. |
| Updating NEW | While a new firmware installs, the screen shows UPDATE MODE, then WRITING with a bar and the per cent. Keep USB in and the FM-1 on until it says DONE. |
| Newer FM-1s NEW | Units sold since autumn 2026 carry another flash chip (UC25HQ80). Since 1.5.1 saves and updates work on it as on the first one. With 1.5 or older on such a unit, install 1.5.1 once through the chip's download mode (the Firmware Hub's Rescue page). |
| Saves | Under a never-ending delay, SPACE on INF or a frozen GRAIN, saves go ahead 10 s after the last note. While a computer records over USB, it saves only after a minute left alone. |

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# Factory sounds and combos

47 sounds with arctic names. Turn PRESETS to step through them by category, then your own.

## Factory sounds

PRESETS

| Control | What it does | Range |
|---|---|---|
| DRONE | Permafrost, Deep Fjord, Cracking Ice, Arctic Hum, Solar Wind, Under the Ice | |
| DRONE · Holvut | Polar Howl, Wendigo, Frozen Scream, Whiteout, Dead Signal, III, Ice Wraith | latched, DRONE on |
| PAD | Polar Night, Snow Veil, Glacier Pad, Midnight Sun, Lichen, Snowdrift | |
| BELL | Ice Bell, Frost Chime, Iceberg Gong, Crystal Harp, Bowed Icicle, Starlight | |
| VOICE | Aurora Choir, Tundra Voices, Blizzard Breath, Northern Call, Empty Horizon | |
| ORGAN | Ice Organ, Base Camp | |
| KEYS | Frost Keys, Snow Globe, Polar Keys, Ice Piano, Snow Owl, Signal Flare, Icicle Pluck, Frost Lute, Northern Light | |
| OTHER | Tundra Strings, Icebreaker, Fog Horn, Svalbard Cello, North Strings | |
| TEXTURE | Wind over Ice | |

## Combos at a glance

Hold + press

| Control | What it does |
|---|---|
| EDIT + OCT− / OCT+ | Undo / redo |
| FX + white keys | PERFORM: FREEZE … STUTTER |
| FX + black keys | GENERATE: T1–T4, MODE, OCTAVE, RANGE |
| SEL (alone) | Chord mode on / off |
| SEL + a knob | Lock the value to the step (sequencer pages) |
| SEL + OCT− | Clear the step's locks (sequencer pages) |
| SEL + OCT+ | The track's LATCH on / off |
| SEQ + white keys | Sequencer tools |
| SEQ + SELECT | Pattern length |
| SAVE + white keys | 1–4 recall, 5–8 save slots A–D |
| GLO + white keys | 1–4 mute, 5–8 solo T1–T4 |
| ARP (alone) | Arpeggiator on / off |
| REC (~1 s) | Clear the track's pattern |
| HOME (alone) | SYSTEM |
| OCT− + OCT+ | Reset CPU PEAK and USB REC DROPS |

Arctic Scream p0larpatch by PolarPatch · the demo, the editor and the cheat sheet: polarpatch.github.io/Arctic-scream-FM-1 · credits and licences

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